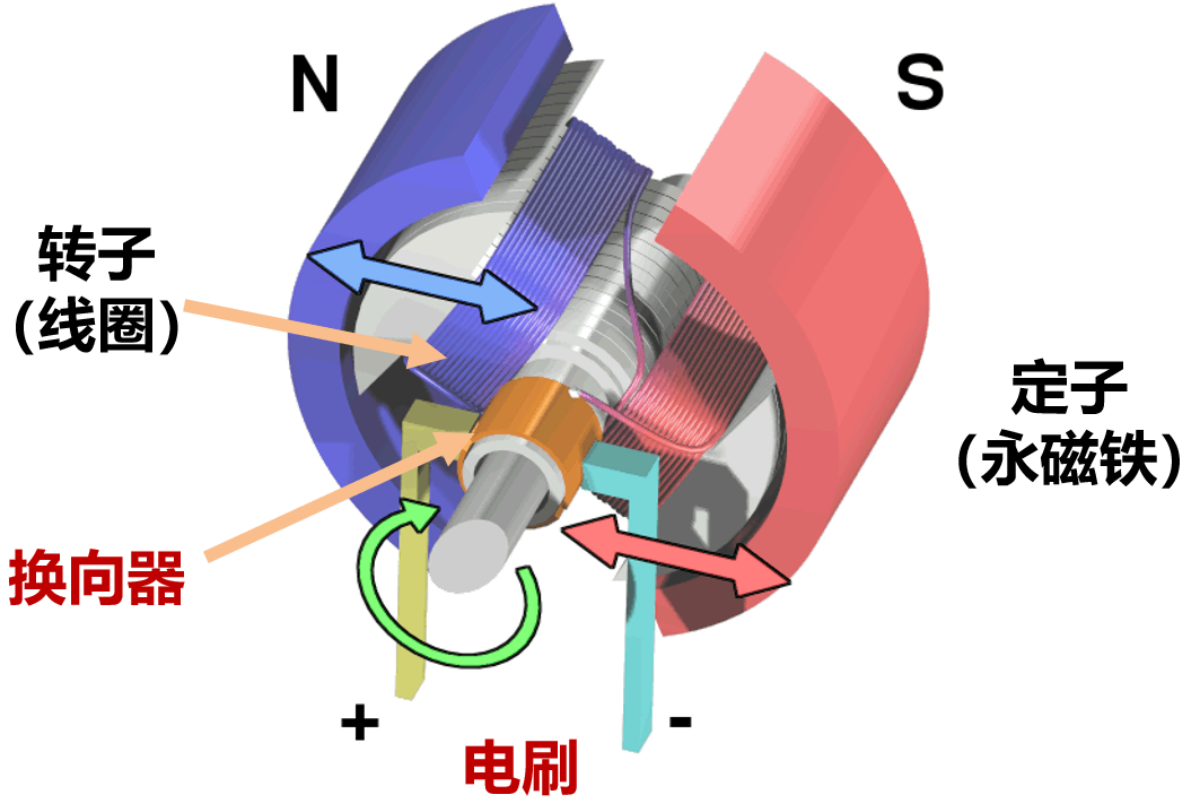
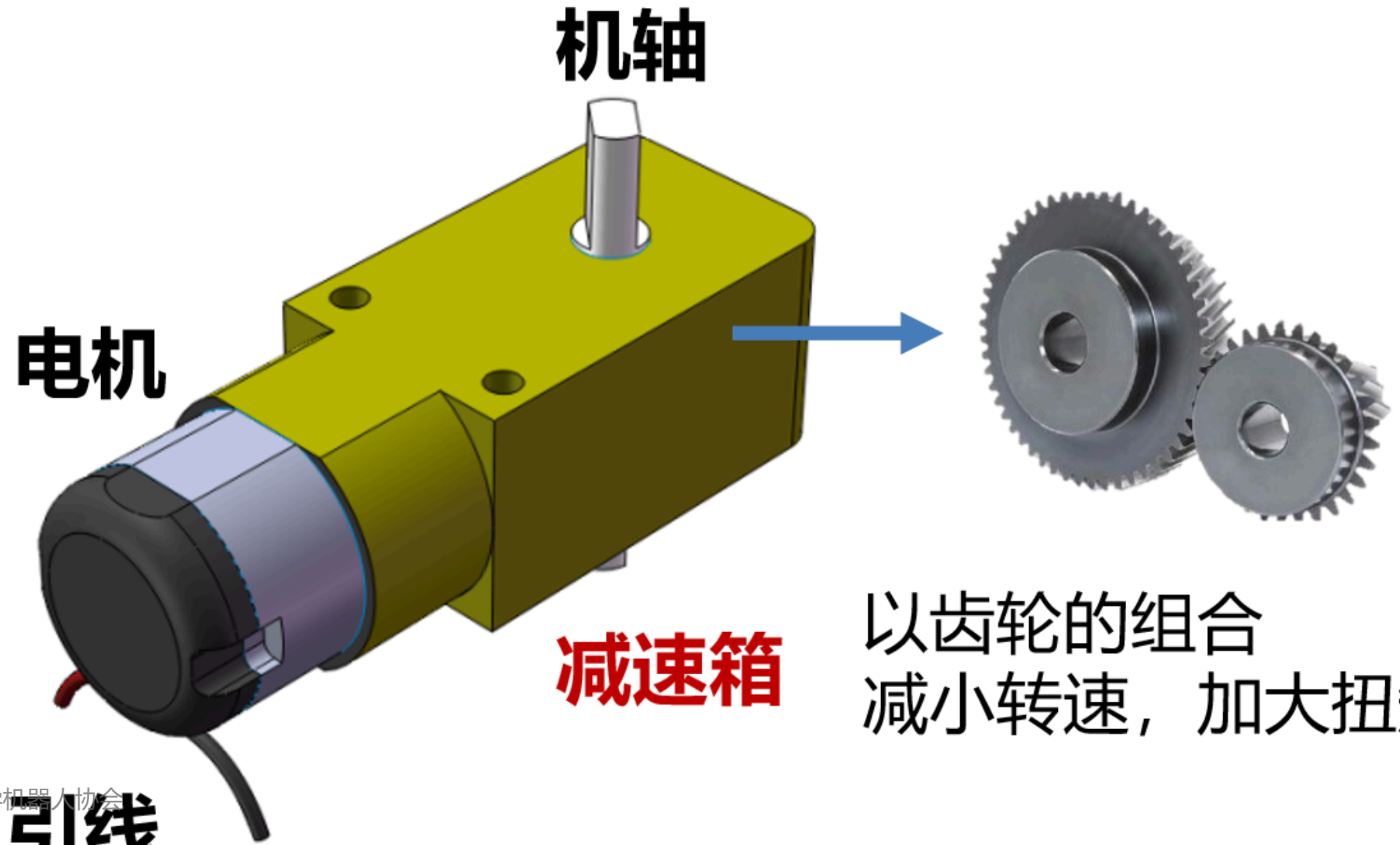
# 电机驱动原理

# 有刷直流电机

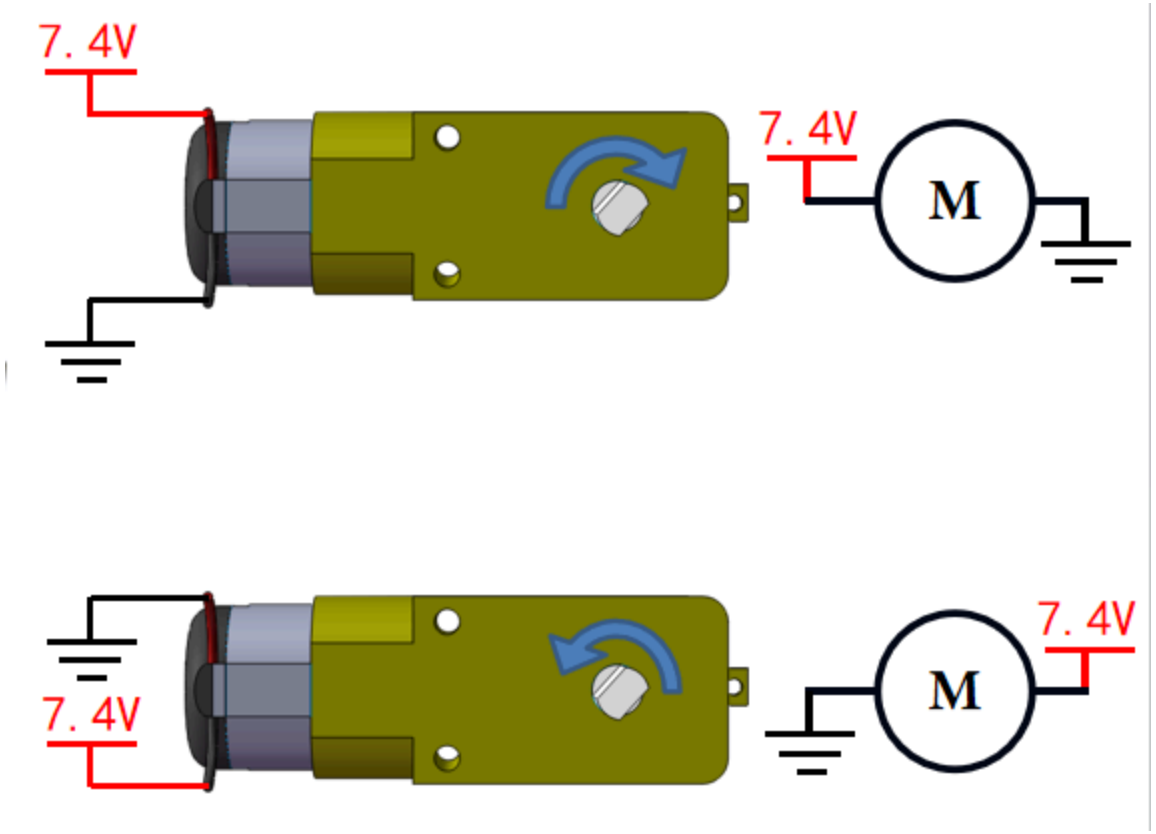
Brushed DC Motor



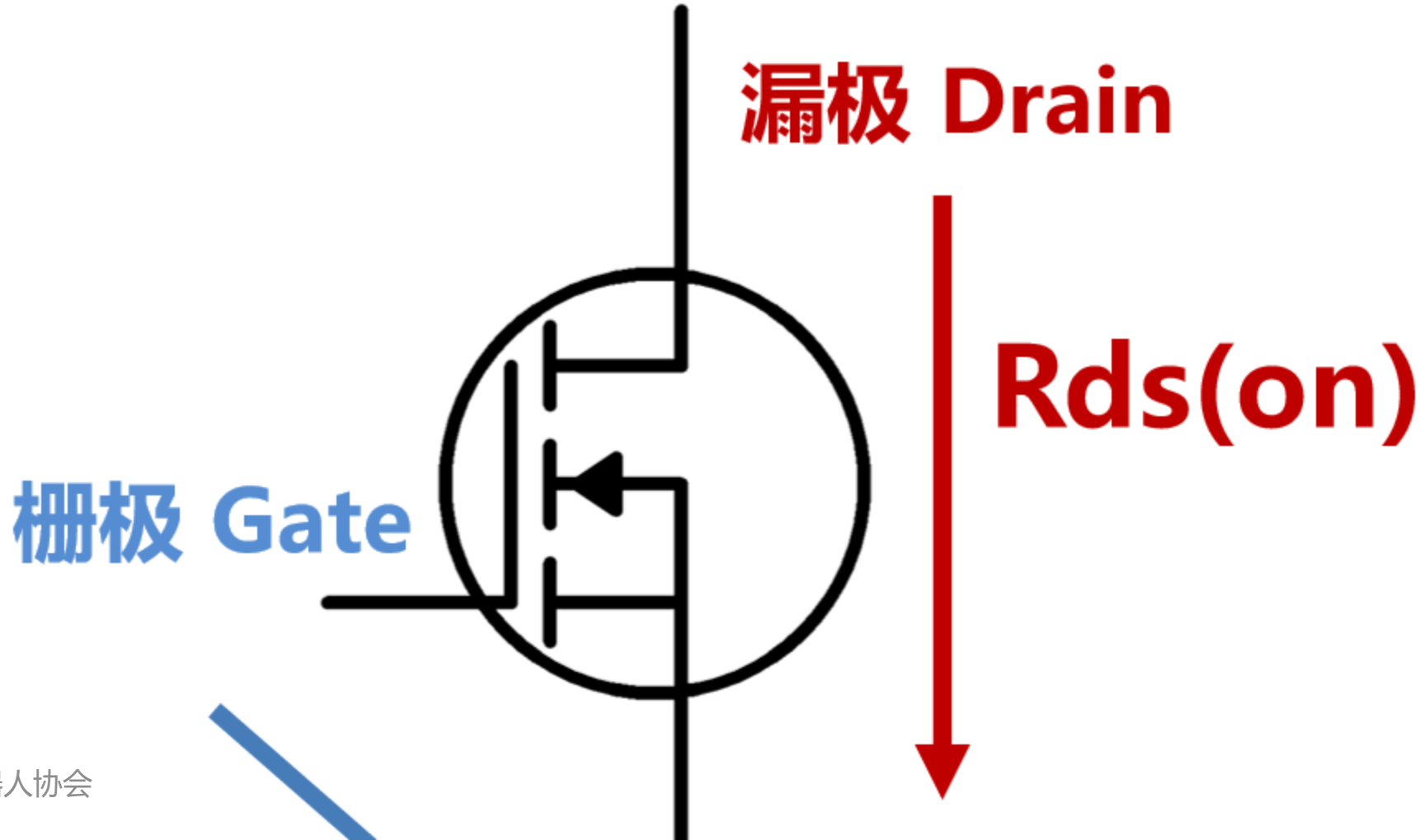



以齿轮的组合
减小转速，加大扭矩

# 直流电机的正反转



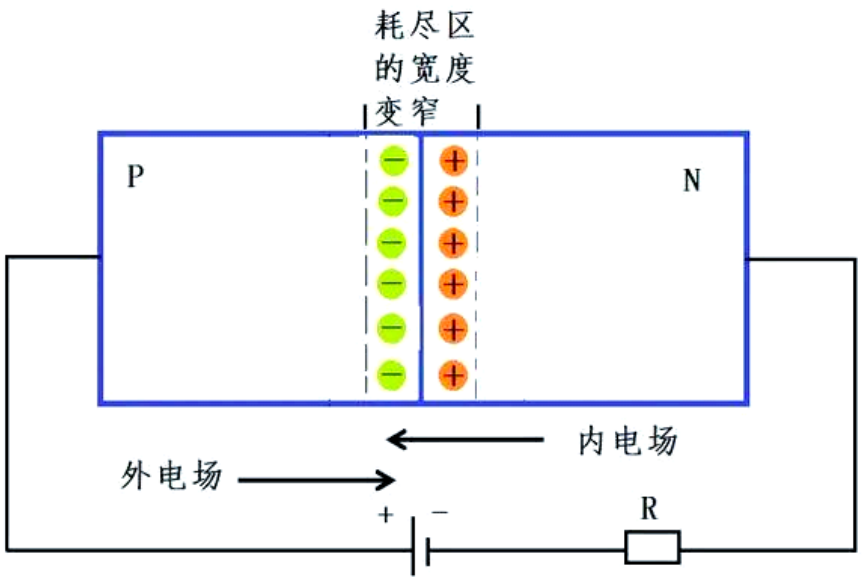
# N沟道增强型场效应管 NMOS

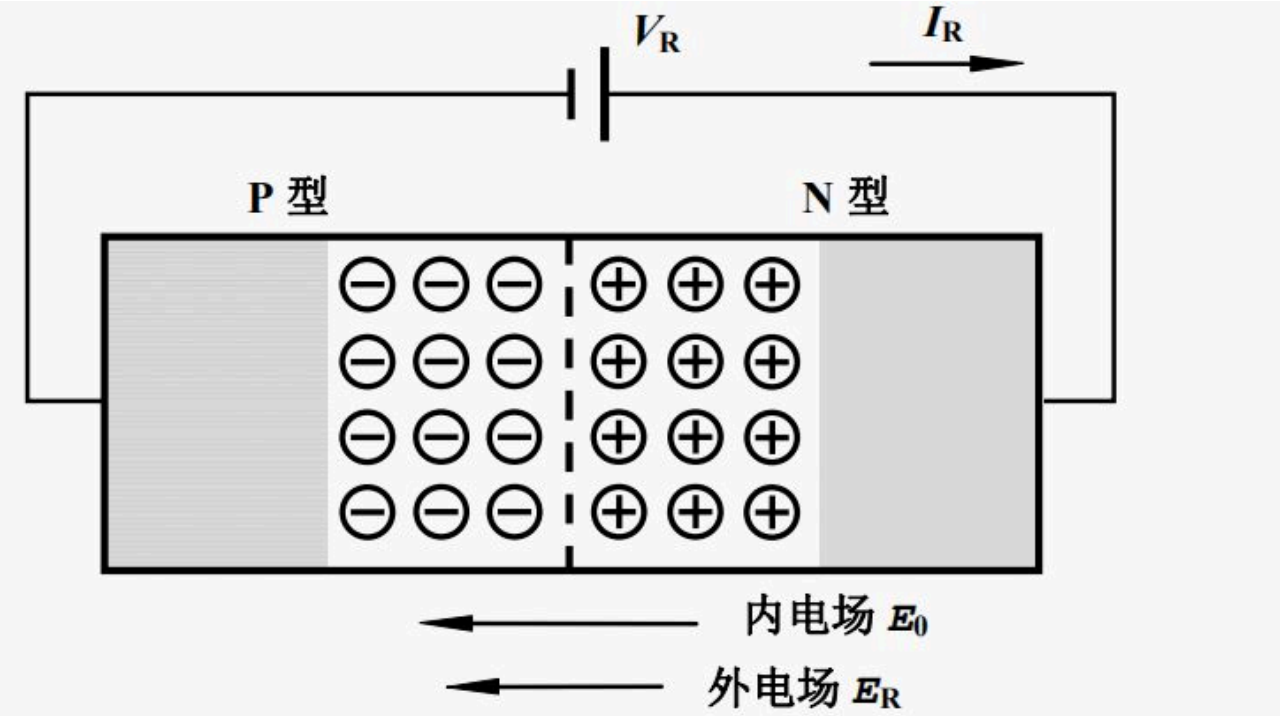


# 拓展：PN结、二极管、BJT和FET

- PN结：半导体、微电子器件的基础
- 二极管：简单的PN结应用器件 单向导通
- 三极管（BJT）：双极型晶体管；分为NPN、PNP；可起开关作用
- 场效应管（FET）：集成电路的核心器件；原理、应用都类似三极管

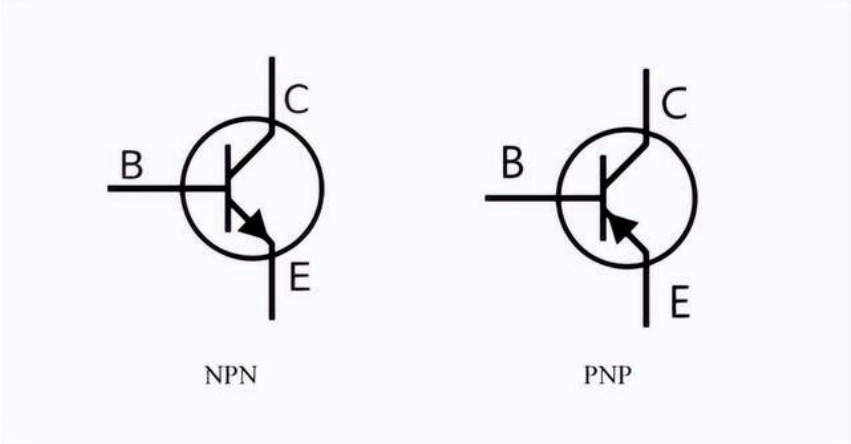
# PN结与二极管

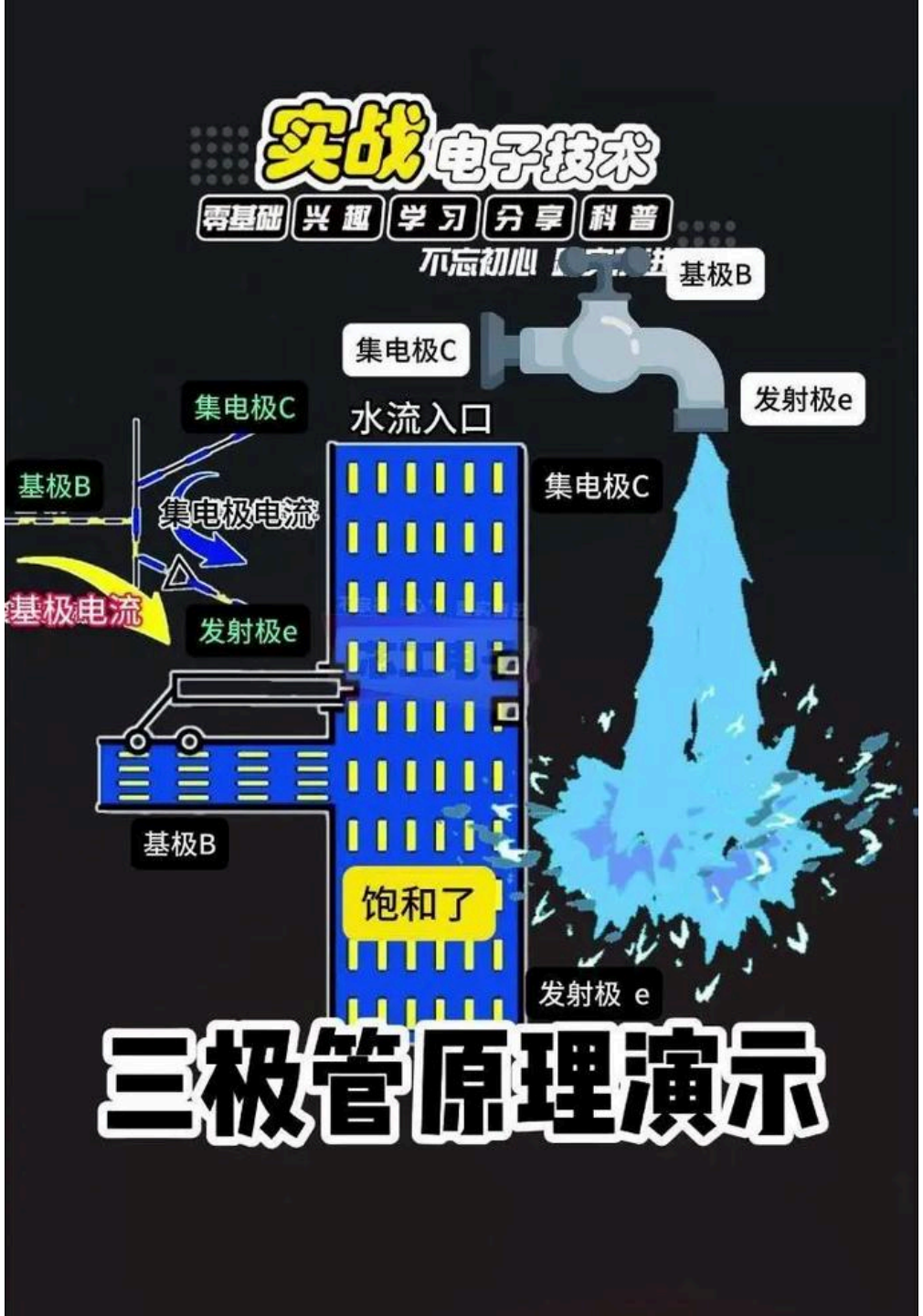


# 三极管

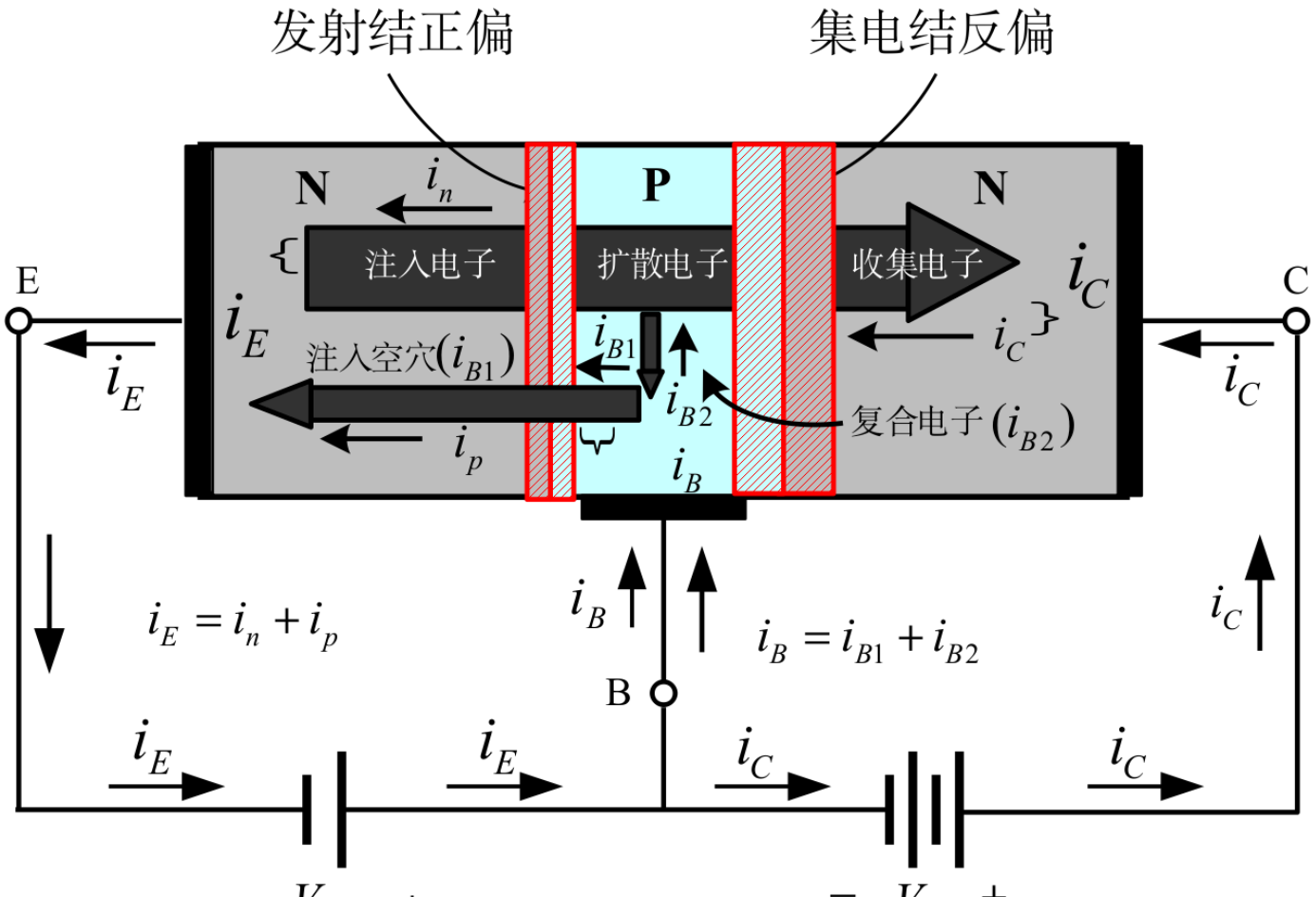
- 全称：双极型晶体管（Bipolar Junction Transistor）
- 分类：NPN、PNP
- 核心原理：像水龙头的阀门一样，控制流过的电流大小



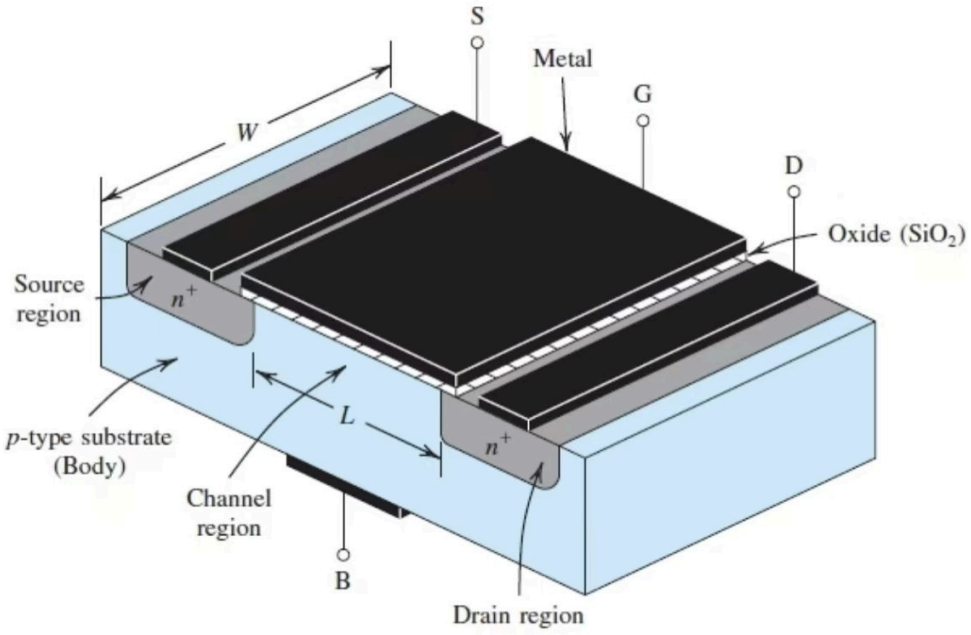
# 三极管原理

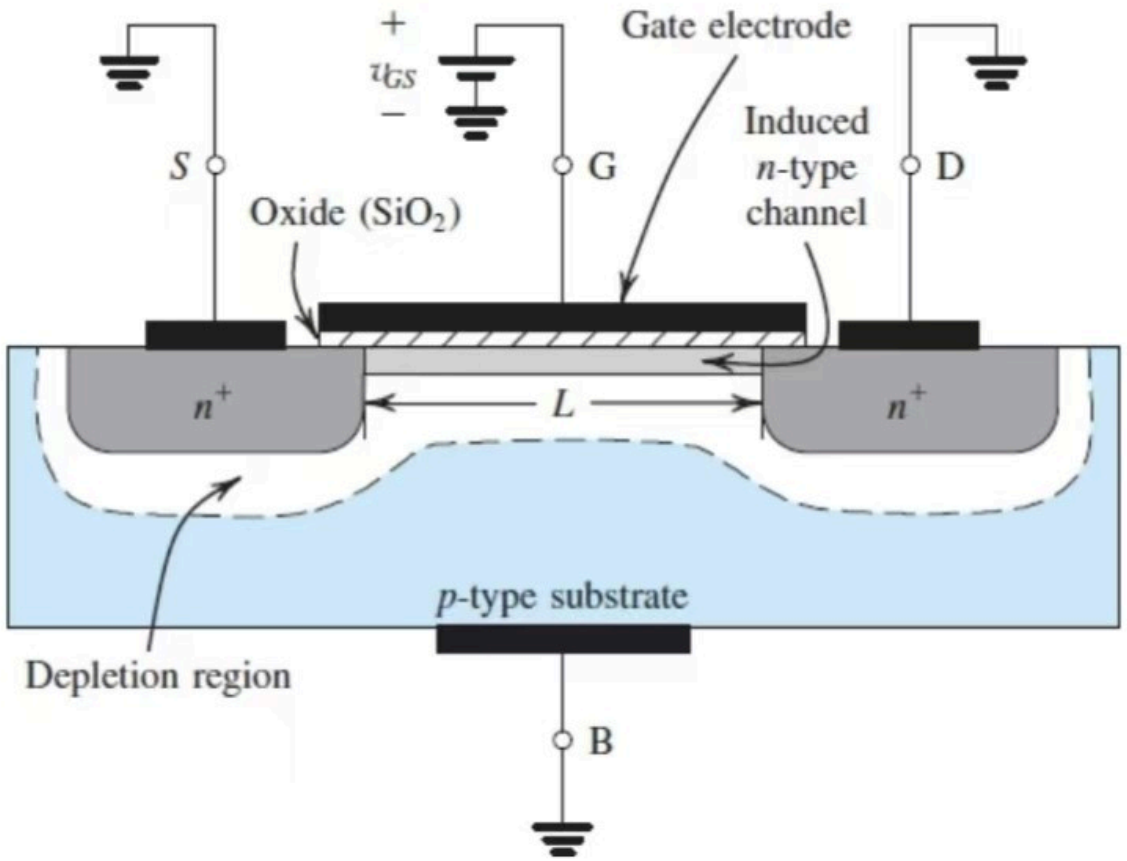
(本图借鉴了电子电路基础chap7的ppt)



# 场效应管（Field Effect Transistor）

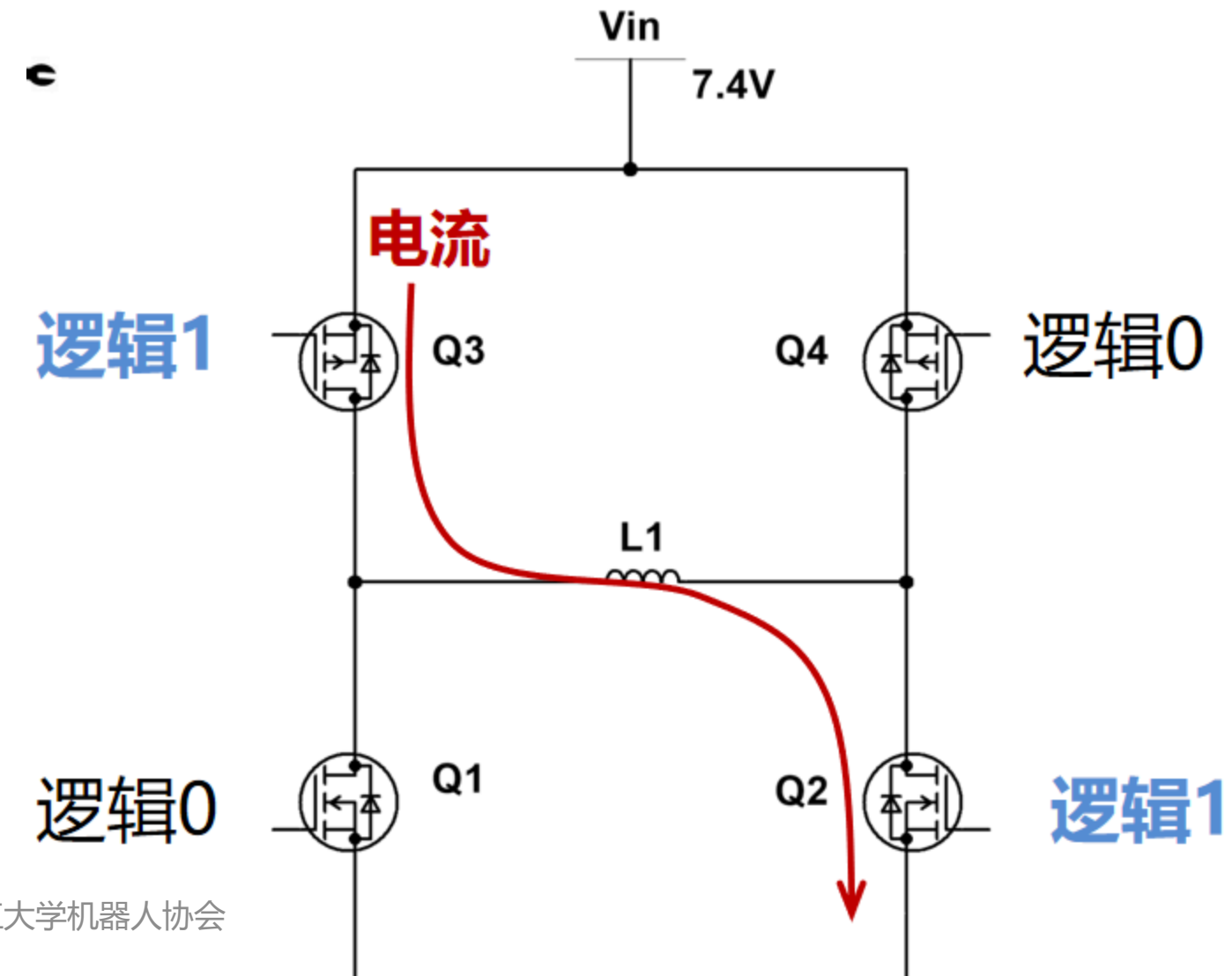
- 集成电路的核心器件；种类样式繁多（MOSFET、FinFET等）
- 以MOSFET为例，下分增强型/耗尽型 NMOS/PMOS
- NMOS可以看作NPN型三极管，PMOS可以看作PNP型三极管
- CMOS:互补型（complementary）金属氧化物晶体管

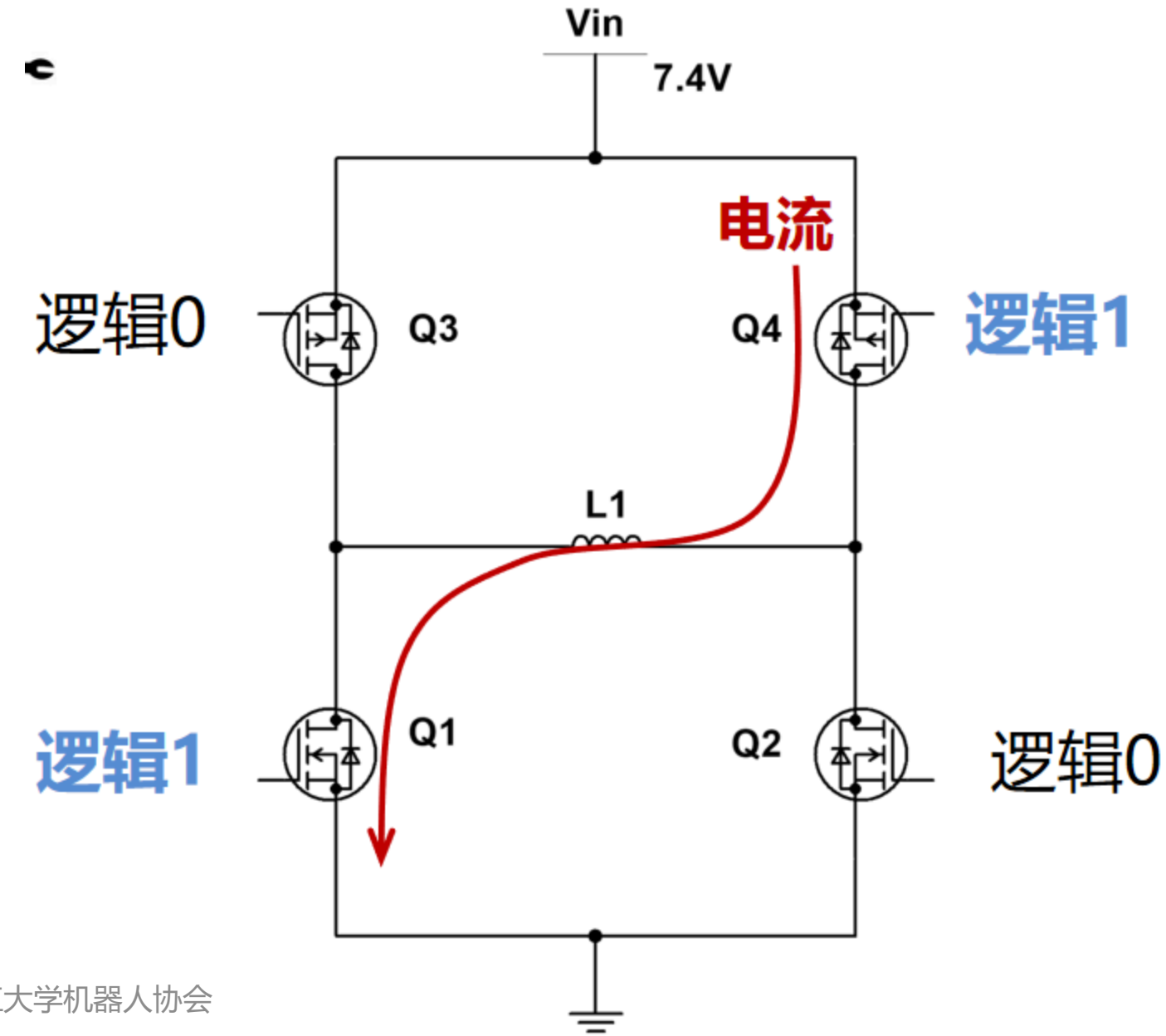

S
Metal
G
W
D
Oxide ($SiO_2$)
Source region
$n^+$
$p$-type substrate (Body)
L
Channel region
$n^+$
B
Drain region

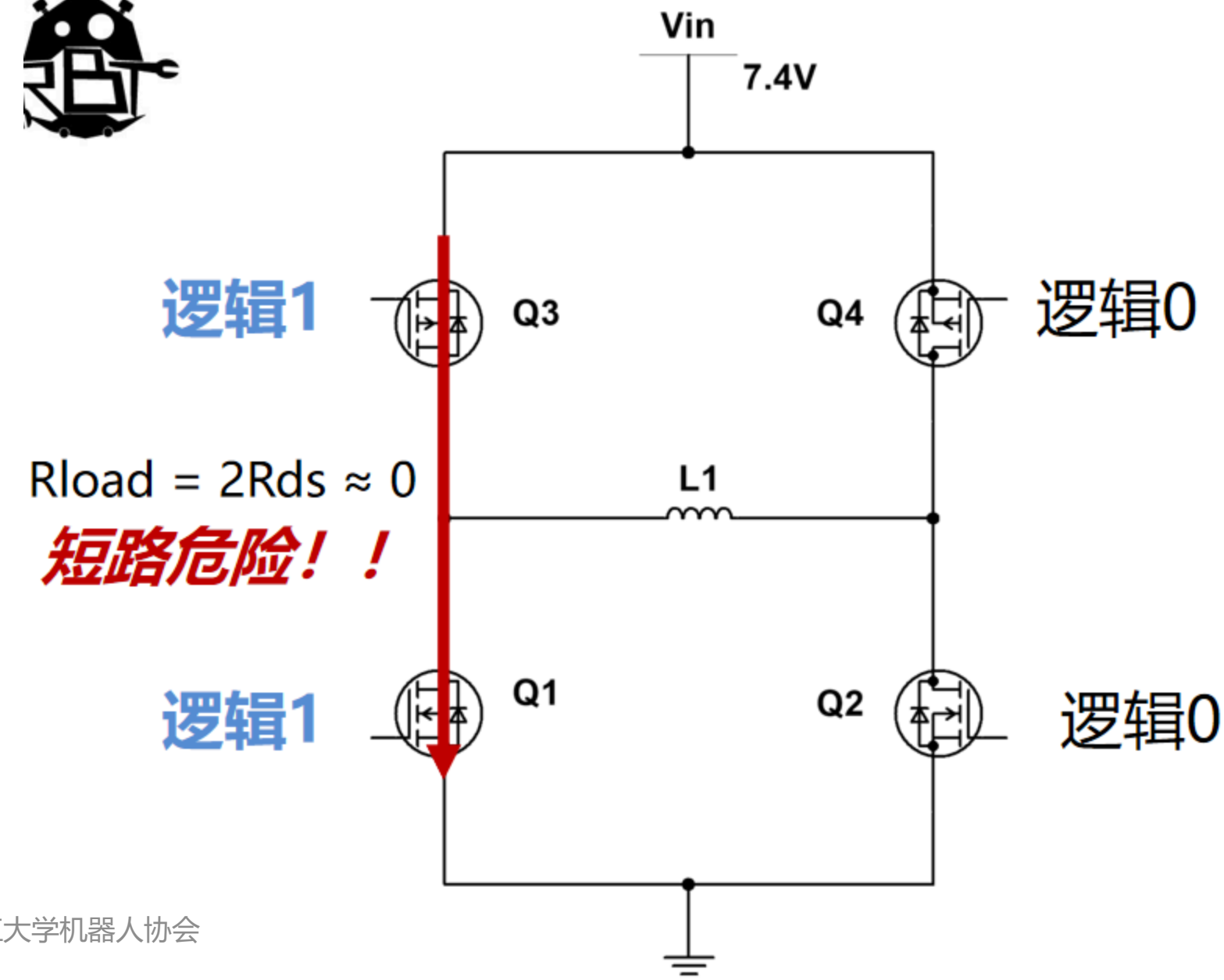

+
$v_{GS}$
−
Gate electrode
S
G
D
Oxide ($SiO_2$)
Induced $n$-type channel
$n^+$
L
$n^+$
$p$-type substrate
Depletion region
B

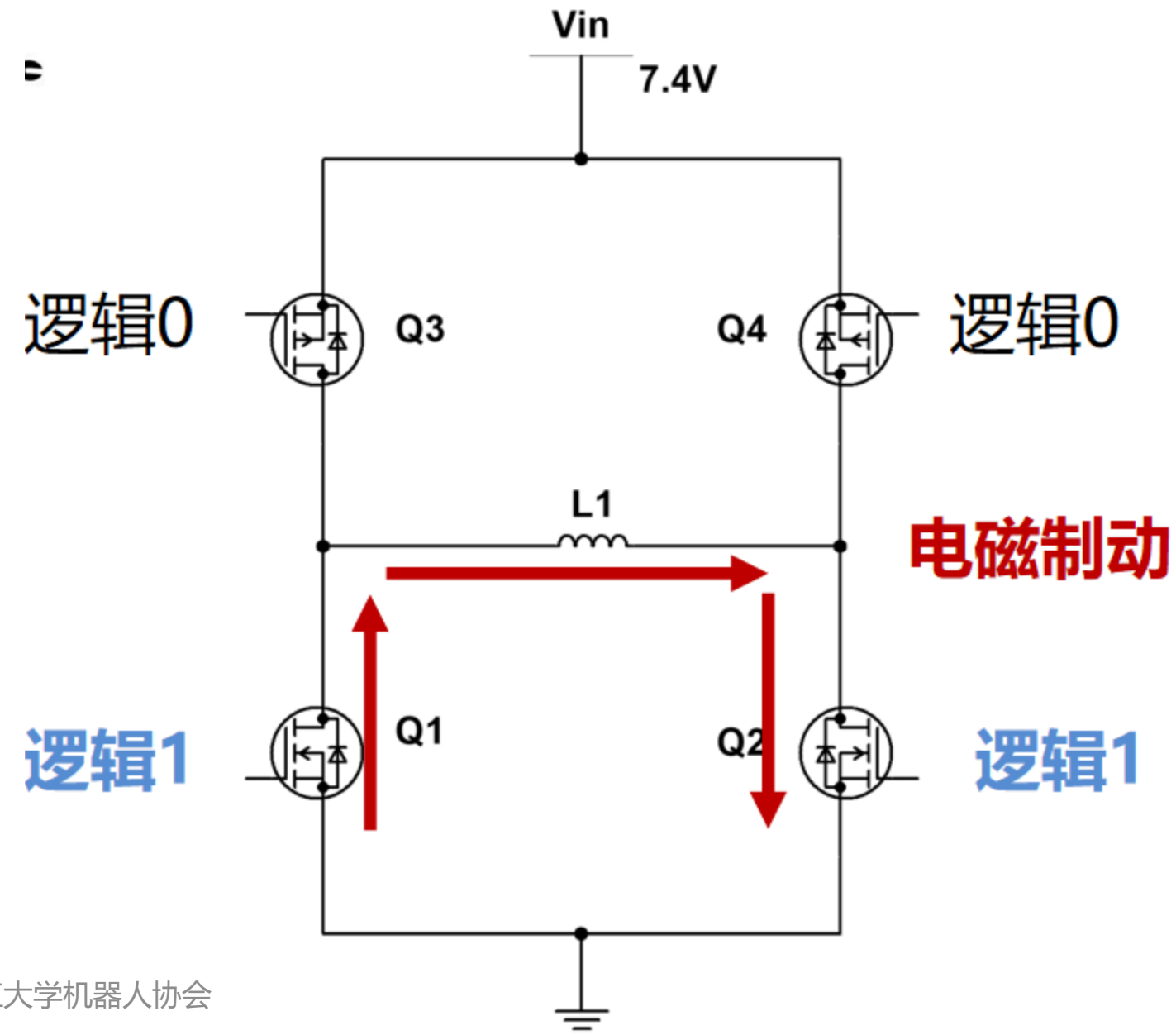
**总而言之，你在这里可以把它当成开关/电磁继电器/水龙头/whatsoever**

# 回归主线

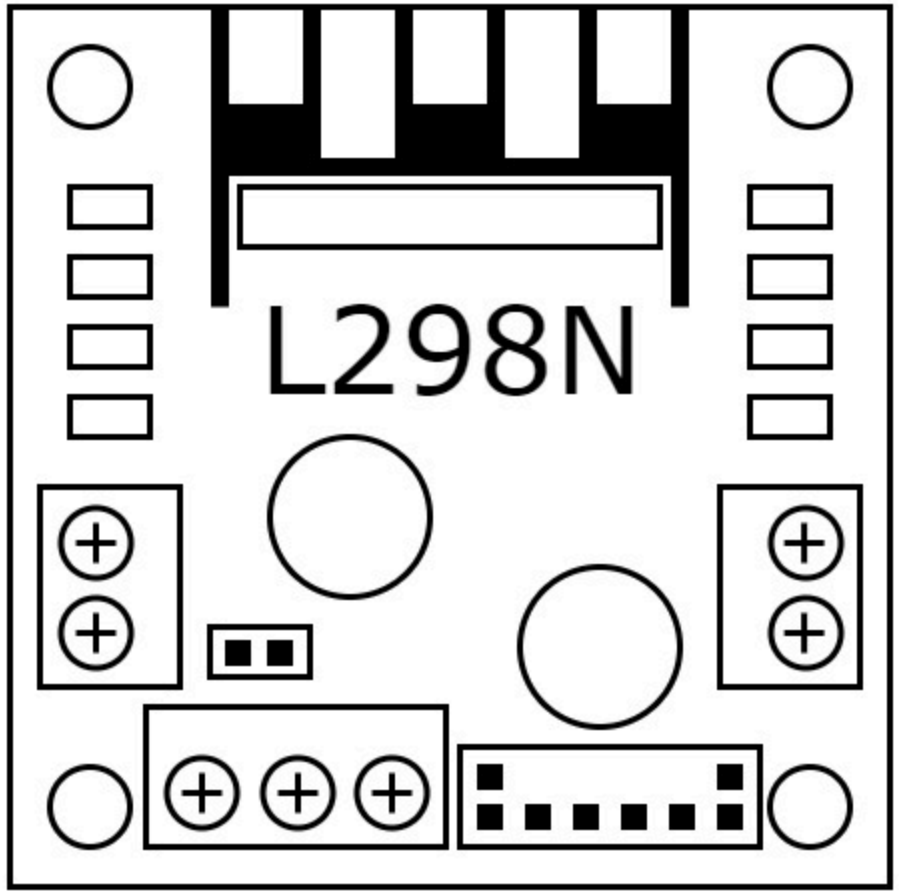


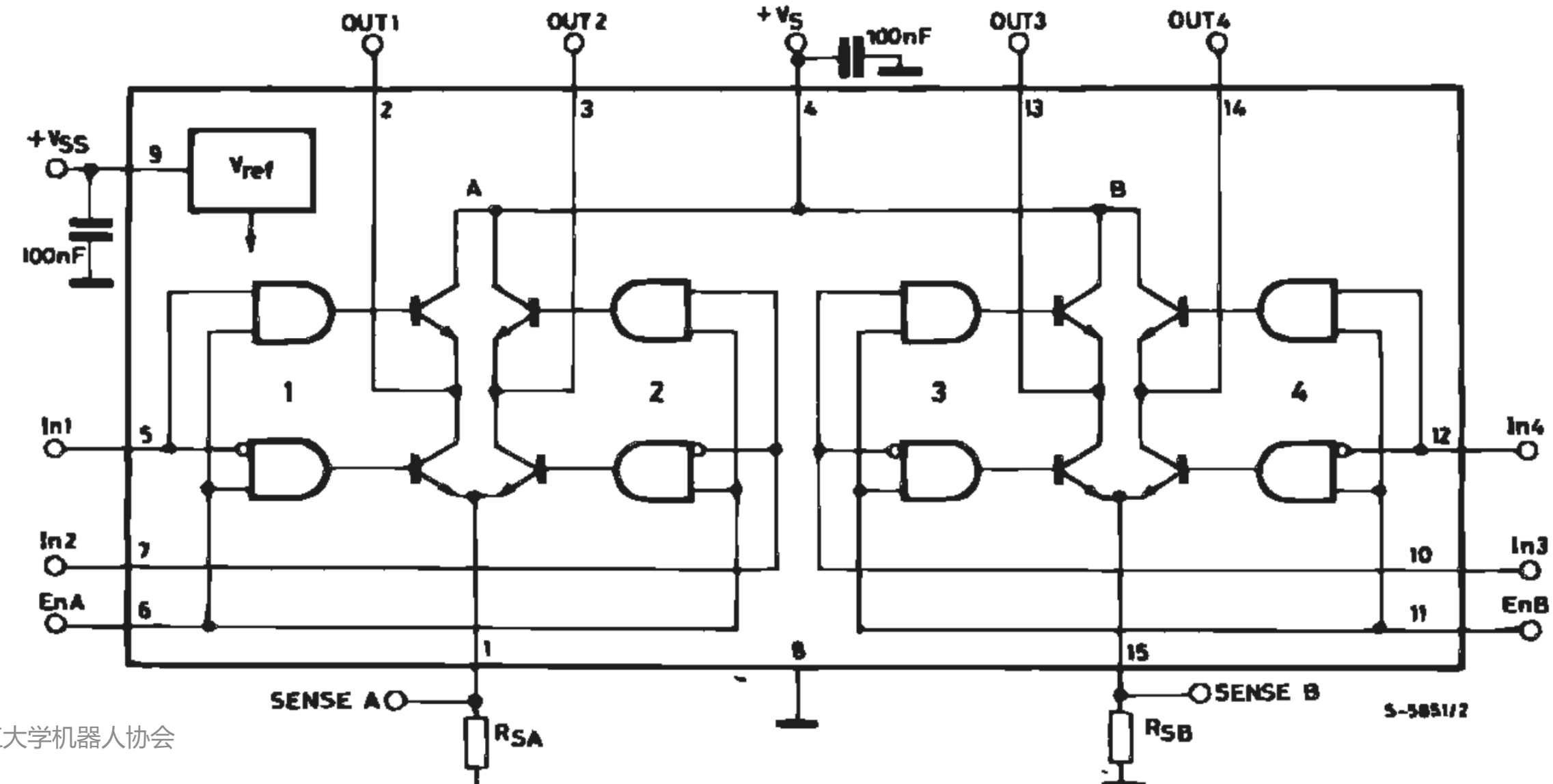

Vin
7.4V
电流
逻辑0
Q3
Q4
逻辑1
L1
逻辑1
Q1
Q2
逻辑0

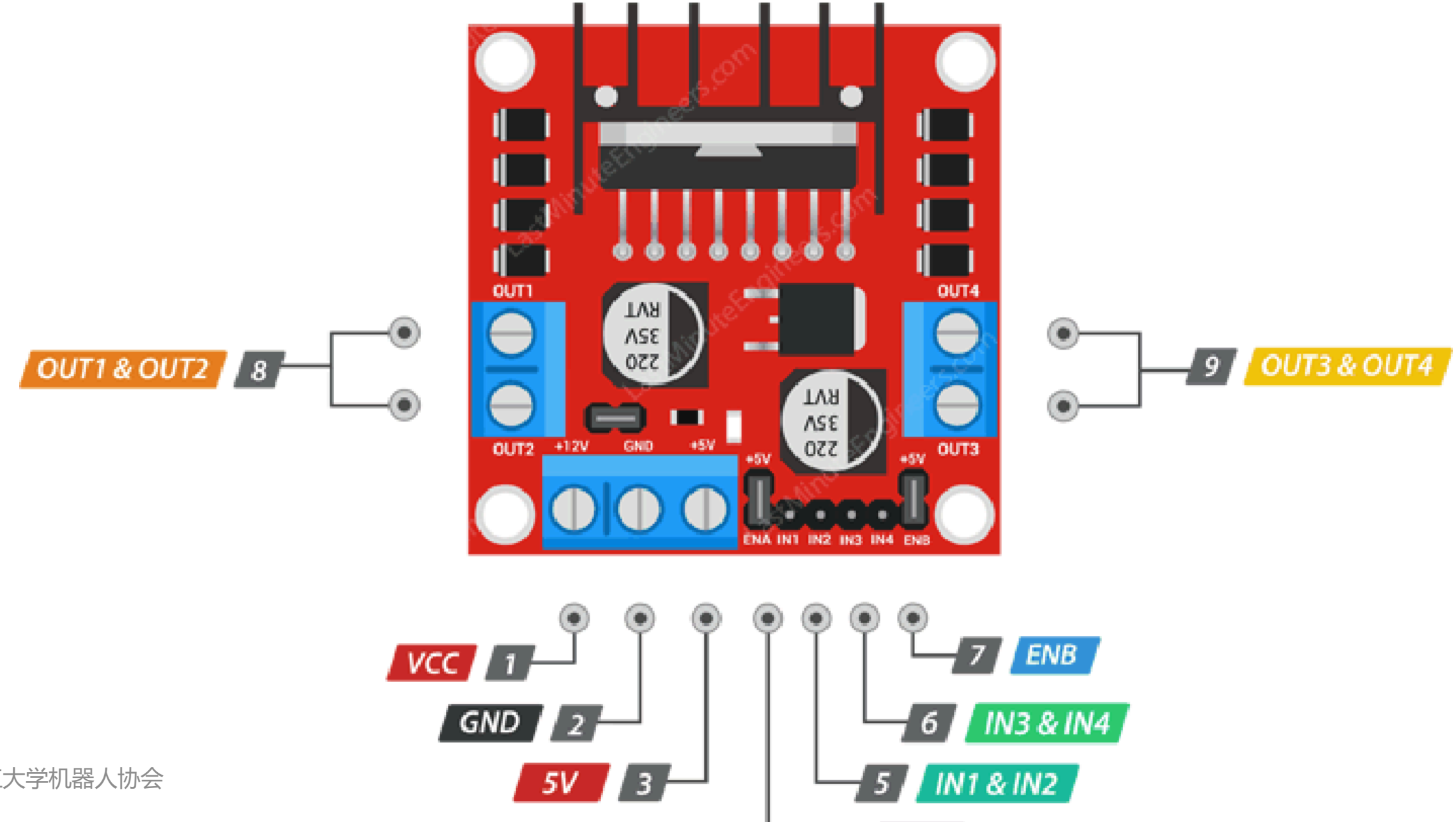

Vin
7.4V
逻辑1
Q3
Q4
逻辑0
Rload = 2Rds ≈ 0
L1
短路危险！！
逻辑1
Q1
Q2
逻辑0

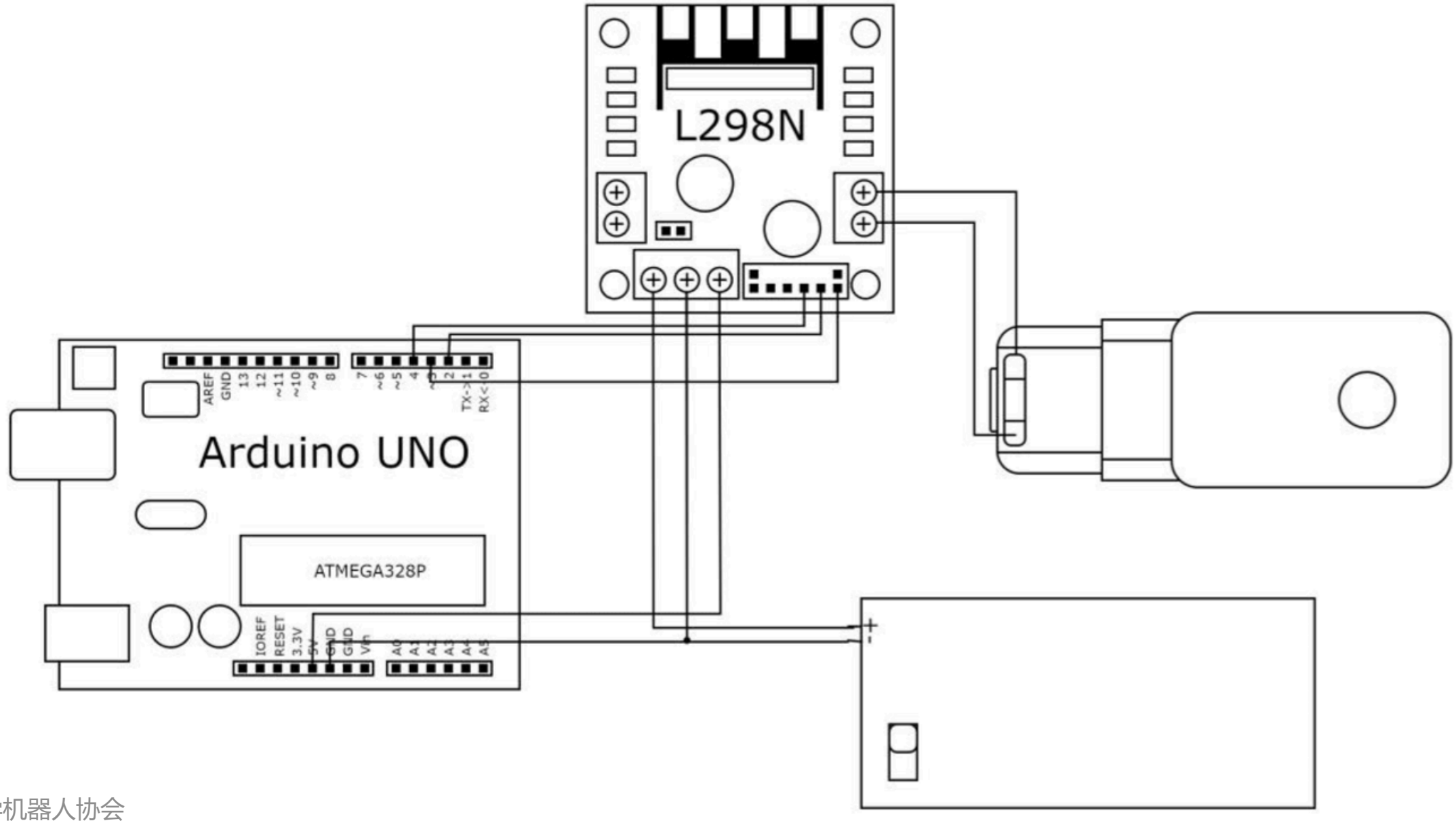

Vin
7.4V
逻辑0
Q3
Q4
逻辑0
L1
电磁制动
逻辑1
Q1
Q2
逻辑1

# 电机驱动——L298N



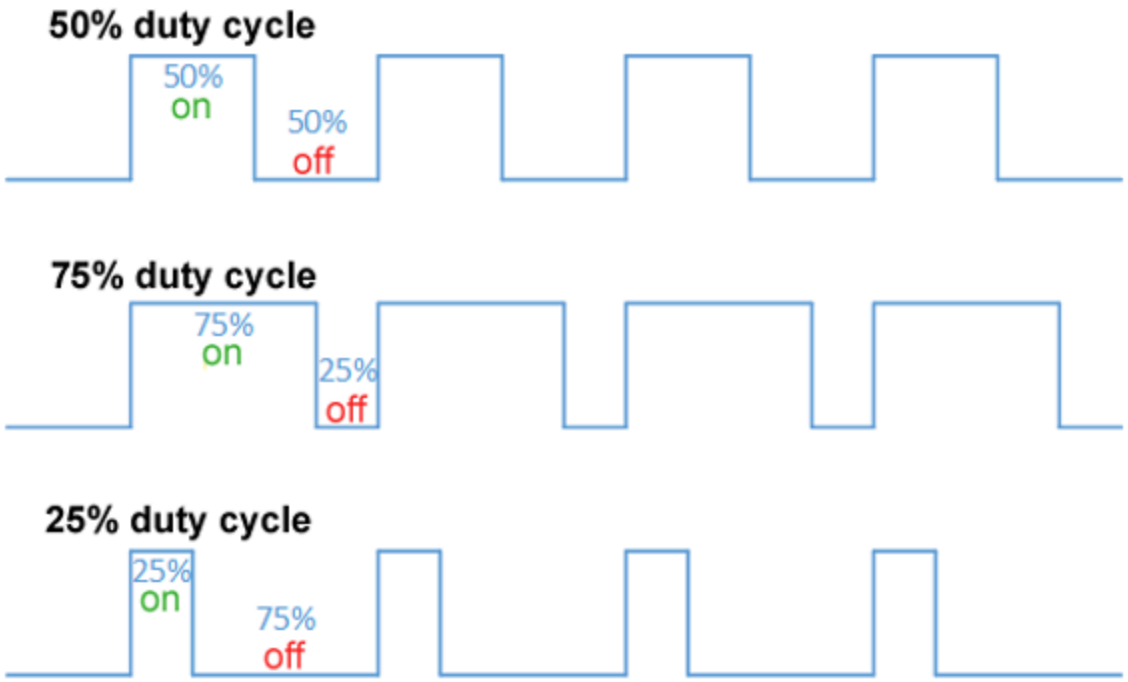

OUT1
OUT2
+VS
100nF
OUT3
OUT4
2
3
4
13
14
+VSS
9
Vref
A
B
100nF
1
2
3
4
In1
5
12
In4
In2
7
10
In3
EnA
6
11
EnB
1
8
15
SENSE A
SENSE B
RSA
RSB
S-5851/2


OUT1 & OUT2
8
OUT1
OUT2
OUT4
OUT3
9
OUT3 & OUT4
+12V
GND
+5V
ENA IN1 IN2 IN3 IN4 ENB
VCC
1
GND
2
5V
3
5
IN1 & IN2
6
IN3 & IN4
7
ENB


L298N
Arduino UNO
ATMEGA328P
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

| EnA | In1 | In2 | 状态 |
|---|---|---|---|
| 0 | - | - | 停止 |
| 1 | 1 | 0 | 正转 |
| 1 | 0 | 1 | 反转 |
| 1 | 1 | 1 | 急停 |
| 1 | 0 | 0 | 滑行 |

ENA =


50% duty cycle
50%
on
50%
off
75% duty cycle
75%
on
25%
off
25% duty cycle
25%
on
75%
off

占空比太低，电机不转?

绕组电流

$$u_L(t) = L\frac{di_L}{dt}$$

$$i_L(t) = \frac{1}{L}\int_0^t u_L dt$$

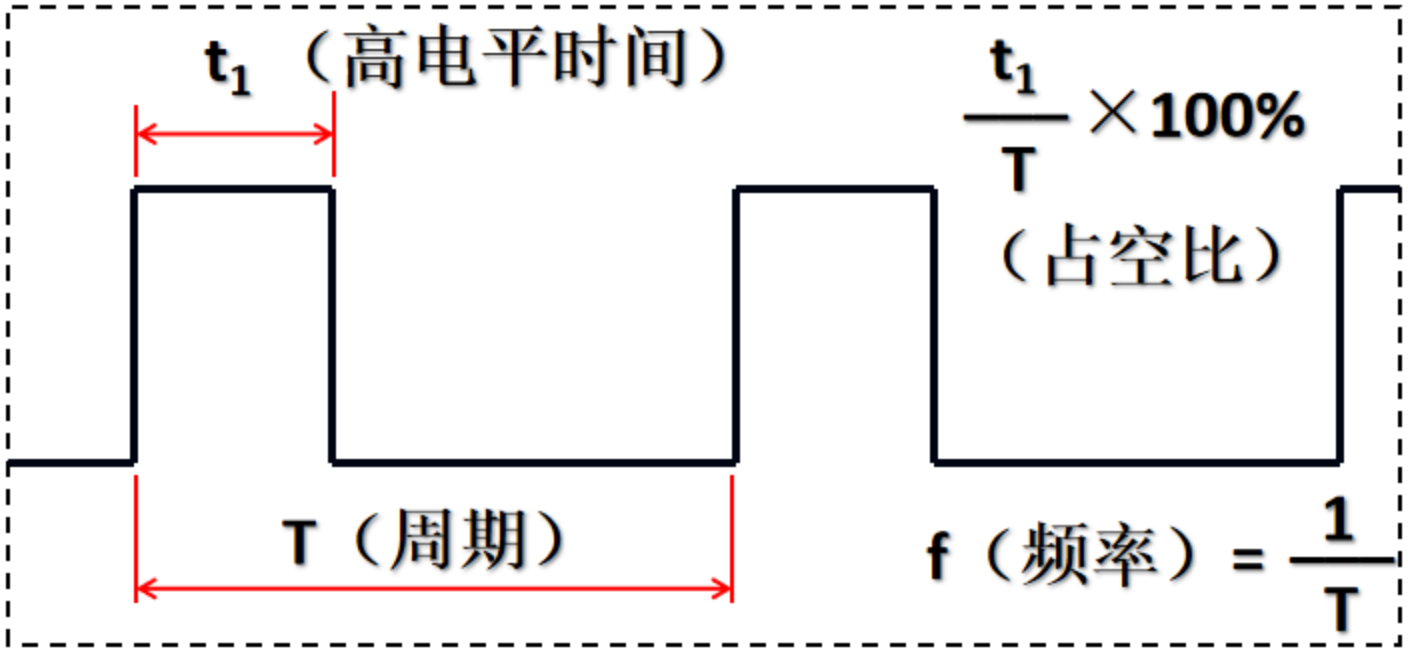
+

磁力矩

$$M = NBI\sin\theta$$

↓

**绕组电感充电时间太短，电流过小**
**不足以克服阻力**

# PWM调速

- PWM（Pulse Width Modulation）脉（冲）宽（度）调制